Dr Deepak Venkateshvaran

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Profile

I lead a laboratory at the Cavendish that measures the mechanical properties of organic semiconductors at the nano and molecular scales, and that treats elasticity as a property that can be dynamically modulated.

Our group holds the highest-resolution stiffness map achieved in an organic semiconductor film, resolved for the first time how stiffness of a molecular crystal arises from the molecular level, and tied the physics of percolation transport to nanoscale mechanical contrast. A decade earlier, my first-author Nature paper connected backbone torsion in conjugated polymers to their charge transport and thermoelectric response, helping to open the field of organic thermoelectrics.

I am a Royal Society University Research Fellow (2021–2029), Fellow and Director of Studies in Physics at Selwyn College, and was seconded to the UK Government in 2024 as a science-policy adviser on energy and advanced materials.

Employment

01/2021 –	**Principal Investigator (Royal Society University Research Fellow)** Cavendish Laboratory, University of Cambridge, Cambridge, United Kingdom
10/2018 –	**College Lecturer in Physics** Selwyn College, Cambridge, United Kingdom
10/2017 –	**Director of Studies in Physics** Selwyn College, Cambridge, United Kingdom
10/2015 – 09/2018	**Isaac Newton Trust Teaching Fellow** Fitzwilliam College, Cambridge, United Kingdom
07/2014 – 12/2020	**Research Associate (Funded through a European Research Council Synergy Grant 2014-2020)** Cavendish Laboratory, University of Cambridge, Cambridge, United Kingdom
10/2008 – 10/2010	**Staff Scientist** (Wissenschaftlicher Mitarbeiter) Walther-Meissner-Institute, Bavarian Academy of Sciences and Humanities, Garching, Germany

Education

10/2010 – 05/2014	**PhD in Physics**, Cavendish Laboratory, University of Cambridge [fully funded through a scholarship] Cambridge, United Kingdom Thesis: *Seebeck coefficient in organic semiconductors*
08/2006 – 08/2008	**Master of Technology in Solid State Technology**, Indian Institute of Technology (IIT) Madras, India CGPA: 9.17/10.00; graduated top of class
06/2004 – 06/2006	**Master of Science in Physics,** Sri Sathya Sai Institute of Higher Learning Prasanthi Nilayam, India CGPA: 5.00/5.00; graduated top of class

Grants

Research Funding as Principal Investigator – over £2.25M since 2021

1.	Royal Society University Research Fellowship (URF) Renewal 2026-2029:	£ 871,571
2.	Royal Society University Research Fellowship (URF) 2021-2026:	£ 722,621
3.	Royal Society Enhancement Expenses 2023, 2024 and 2025:	£ 297,758
4.	Royal Society Enhancement Expenses 2021/2022 and 2022/2023:	£ 243,828
5.	Costed Extension for the Royal Society Policy Associate Scheme 2026:	£ 30,255
6.	Wiener-Anspach Foundation Research Grant 2024/2025:	€ 30,000
7.	Royal Society Additional Research Expenses 2021, 2022, 2023:	£ 17,000
8.	Royal Society Research Grant 2021:	£ 24,948

Industrial Support

9. Park Systems UK: equipment and personnel support to the laboratory, including a £ 47,000 equipment subsidy in 2021

Large-Scale Research Facilities

10. Co-Investigator (2026): ISIS Neutron and Muon Source Beam Time; Instrument: TOSCA. 24 hours awarded. Proposal RB2620625: "The impact of isotopic substitution on the vibrational spectrum of molecular organic semiconductors"

Scholarships and early-career awards – approximately £100,000

11. Overseas PhD Scholarship from Cambridge Commonwealth Trusts 2010-2013:	£ 75,000+
12. Isaac Newton Trust Teaching Fellowship:	£ 15,000
13. Microsoft grant to attend the Lindau Nobel Laureates Meeting 2012:	£ 5000
14. Selwyn College Academic Grants:	£ 5000+
15. Fitzwilliam College Travel Grants:	£ 1000
16. IIT-DAAD Scholarship 2007/2008:	€ 10,000
17. IIT Madras Teaching Fellowship 2006/2007:	₹ 60,000

Awards

Fellowships and Appointments

Royal Society URF (2021 – 26, renewed 2026 – 2029)
Visiting Professorship, Institute of Science Tokyo, Japan (2026) – funded by the World Research Hub Program
Nominee, Phyllis Beddington Wiener Visiting Chair Professorship, ULB, Belgium (outcome 2027)
Royal Society Science Policy Secondment at the UK Government (2024 – 2025)
Fellowship, Selwyn College, Cambridge (2018 –)
Bye Fellowship, Selwyn College, Cambridge (2017 – 2018)
Isaac Newton Trust Teaching Fellowship, Fitzwilliam College, Cambridge (2016 – 2018)
Bye Fellowship, Fitzwilliam College, Cambridge (2015 – 2018)

Prizes and Medals

Rank Prize *Seeing in Science* (2021)
Lundgren Research Award (2013)
Institute Merit Prize and Medal, IIT Madras (2008)
Dr G Venkataraman Gold Medal, Sri Sathya Sai Institute of Higher Learning (2006)

Selected early-career distinctions

UK representative at the 62nd Lindau Nobel Laureate Meeting for Physics (2012)
Fitzwilliam College Senior Scholarship (2012)
Deutscher Akademischer Austausch Dienst (DAAD) Fellowship (2007 – 2008)
Wilhelm und Heraeus-Stiftung Travel Grant (2008)
Council of Scientific and Industrial Research Fellowship (CSIR-UGC JRF NET) – Qualified (2006)
Rank 56/4904 in all India Graduate Admission Test in Engineering (GATE) for Physics (2006)
Indian Academy of Sciences Summer Research Fellowship (2005)

Talks and Conferences

Delivered keynote talks at Park Systems' Nano Scientific Forum Europe (NSFE) in 2022, 2023 and 2026.
Delivered invited talks at major global conferences: MRS Seattle, MRS Boston, SPIE San Diego, Spintech, SpinOS, and the Annual Meeting of the Japanese Physical Society.
Delivered invited seminars at world class research institutions such as KTH Stockholm, University of Tokyo, TU Delft, TU Munich, Oxford University, Queen Mary University, IIT Madras, and the Max Planck and Jozef Stefan Institutes.
A current record of international speaking engagements is at https://deepak-venkateshvaran.github.io/talks/

Conference Organisation: Co-organiser of the annual ULB-Cambridge-RIKEN workshop *Organic Semiconductors for the Next Era* (2025, 2026). Session chair at NSFE 2023, MRS Seattle 2024, ECME 2025. Organising committee at ECME 2025.

Teaching

I co-lecture an MPhil course (NE.10) on Energy Harvesting (Thermoelectrics, Pyroelectrics, Piezoelectrics) at the Department of Materials Science and Metallurgy at Cambridge University and a Part III Physics course (IDP3) at the Cavendish Laboratory on the theme of Energy Science (Hydrogen Technologies). I've taught undergraduate courses within the Part 1B, Part II, and Part III NST Physical Sciences at Cambridge University. I've co-supervised three master thesis projects, six doctoral projects, and three postdoctoral projects within the Physics department.

As Director of Studies in Physics, Selwyn College has risen over my tenure from the middle of the Cambridge college rankings for academic performance into the top three in recent years.

Management Training – Innovation and the Business of Science

02/2023 – Introduction to Management – Imperial College Business School Executive Education
03/2023 – Leadership Effectiveness – Imperial College Business School Executive Education
03/2023 – Royal Society Science Policy Primer – Introduction to Government Policy and Science based Decision Making

Science Policy in Government

I've worked on policy within the Department of Science Innovation and Technology (DSIT) of the UK Government. In 2024, my position was organised through DSIT's expert exchange secondment scheme and funded through a Royal Society science policy associate scheme. Salient responsibilities of mine in government were as follows:

- I led a team to develop a policy report analysing the last 20 years of funding for graphene research and innovation within the UK, including successes, failures, and lessons learned.
- I've written policy papers on energy harvesting and advanced materials technologies for DSIT
- I've liaised with UKRI on government funded programs for energy materials and energy harvesting; represented government in visits to UKRI-funded companies such as Paragraf and Levidian; reviewed documents and policy briefs submitted by the Henry Royce Institute to the UK Government; engaged with external bodies (ScotChem Ltd) to identify emerging themes and developments on advanced materials for the UK's National Materials Innovation Strategy

International Research Collaborations

I've built a global consortium of academics working on themes that bolster research activities on nanomechanics of energy materials. Within this consortium, I conceive and direct research projects, including the supervision of PhDs and postdocs both in Cambridge and within our partner institutions.

Our partners together with their contributed theme of expertise:

1. Prof Per Claesson at KTH Stockholm, **Sweden** (Nanomechanical Measurement Techniques)
2. Prof Yoann Olivier at University of Namur, **Belgium** (Micromechanical Molecular Dynamics)
3. Dr Guillaume Schweicher at ULB, **Belgium** (Chemistry and Devices of Molecular and Chiral Organic Semiconductors)
4. Prof Erin R Johnson at the Dalhousie University, **Canada** (Density Functional Theory of Crystal Mechanics)
5. Prof Taishi Takenobu at Nagoya University, **Japan** (Organic Polymer Semiconductors)
6. Prof Kazuo Takimiya at RIKEN, **Japan** (Next Generation High Mobility Organic Crystals)
7. Dr Ulrike Kraft at the Max Planck Institute in Mainz, **Germany** (Electrolyte Gated Organic FET Devices)
8. Park Systems **UK** (AFM Instrumentation)

Leadership in Academia: Impact, Management, Service, Outreach

Impact

- My PhD work at the Cavendish Laboratory (2010 – 2014) spun out multiple new projects, either completed or ongoing. Over 10 PhDs and Postdocs were hired at the Cavendish Laboratory based on the outcomes of my PhD in

organic thermoelectrics. The intellectual property, I understand, has generated several million pounds in follow-on research funding for Cambridge University

- I setup two new laboratories at Cambridge University: a Nanomechanics laboratory (2021 – 2022) and, together with scientists at Hitachi Cambridge Laboratories, a Spintronics Laboratory (2014 – 2015)

People Management

- Trained and supervised multiple Master, PhD, and Postdoc projects at the University of Cambridge (2014 –)
- Organised and lead research activity between Cambridge University, ULB Belgium, and KTH Stockholm (2021 –)
- Trustee and Governing Body Member, Selwyn College, Cambridge (2018 –)
- Coordinated regular meetings between academics working on a 10-million-euro ERC Synergy Grant (2014 – 2021)

Service

- Selection Committee Member – Henslow Research Fellowship at Cambridge University (2026, 2022, 2018); Trevelyan Research Associates at Selwyn College Cambridge (2019 –); Gavin Boyle Fellowship in Cosmology and Exoplanetary Research (2019)
- Panel Member – Cavendish Laboratory Graduate Student Funding Committee (2023 – 2026)
- Assessor – Undergraduate, Masters, and PhD theses at Cambridge University (2014 –)
- Review grant proposals for National Research Funding agencies in four countries: Global Excellence Initiative, Canada (2025), National Science Center, Poland (2025), Czech Science Foundation, Czech Republic (2021), Ser Cymru II Rising Stars Fellowship, Wales (2017)
- Review for *Nature*, *Nature Communications*, *Science Advances*, *Advanced Materials* amongst other journals

Outreach

- Member of the Access Committee at Selwyn College
- Delivered talks to high school students on open days, and in financially deprived areas of England's North
- Delivered talks to the public on the physics of Indian Drums, and separately, on organic electronics
- Written articles on the Physics of Indian Drums and on organic electronics for the Selwyn College Calendar

Patent and Machine Learning (ML) Certification

Artificial Synapse - Patent with Hitachi Cambridge Laboratories (published in March 2021)
Antiferromagnetic memristor based synapses for neuromorphic computing – John Armitage, Deepak Venkateshvaran, Joerg Wunderlich (European Patent Application EP19199322A)

My machine learning training (Coursera/Stanford, edX/Harvard, DeepLearning.AI, 2016-2020) support a doctoral project I supervise on machine vision applied to Atomic Force Microscopy nanomechanical data and underpins new project modules linking nanomechanics to the Department's Data Intensive Science programme.

Physics and Music

I have played the Indian Tabla for over 30 years, have contributed to three studio-recorded music albums, and have composed scores for modern art films that premiered in the UK. I lecture publicly on the instrument's acoustics – the tabla's loaded membrane draws its overtones into a harmonic series, and that same physics is now the basis of a device my laboratory is building from organic semiconductor films.
Public lecture on the physics of the tabla: https://youtu.be/SOlPKuPtFfs

Top 5 peer-reviewed publications including a description

* indicates corresponding author, † indicates joint first authors

1. Ki-Hwan Hwang, D. Brandt … Deepak Venkateshvaran*, **Nature Communications 17**, 1621 (2026) **URF Work**

   The mechanical properties of molecular semiconductors bear contribution from the chemical structure of molecules in the lattice as well as from intermolecular interactions. The contribution to stiffness from the chemical structure proves subtle and difficult to detect. In this work, we demonstrated how such a subtle effect can be quantified in nanoscale measurement.

2. Illia Dobryden, V. V. Korolkov … Deepak Venkateshvaran*, **Nature Communications 13**, 3076 (2022) **URF Work**

   We demonstrated the highest resolution nanomechanical property measurement in an organic polymer using acoustic frequency comb generation and intermodulation techniques. We combined it with a technique called Higher Eigen Mode (HEM) imaging for sub-nanometer spatial resolution of organic polymers under ambient conditions. This work contributed to changing the narrative around the nanostructure of the high mobility polymer IDTBT, from being amorphous to having nanosized domains of order. It complemented the work of Prof Alberto Salleo at Stanford University (ACS Macro Lett. 2021, 10, 10, 1306–1314) and was given significant visibility at four international conferences between 2023 and 2024. It was a collaboration between our group in Cambridge, our industrial partner Park Systems UK, and the surface science group of Prof Per Claesson at KTH in Stockholm.

3. Shu-Jen Wang†, Deepak Venkateshvaran†* *et al.*, **Nature Electronics 2**, 98-107 (2019) **Postdoctoral Work**

   We tried to establish the exchange-coupling mechanism behind long spin diffusion in doped conjugated polymers and documented – with more candor than is usual – the routes to all-electrical spin injection in organic semiconductors that do not work, and why. The negative results were continued in a follow-on paper, Physical Review Materials 6, 024601 (2022), and have saved other groups considerable time.

4. Deepak Venkateshvaran *et al.*, **Nature 515**, 384–388 (2014) **PhD Thesis Work**

   This work constituted the crux of my PhD thesis. When I arrived in Cambridge, I was fully funded by the Cambridge Commonwealth Trusts. My PhD supervisor, Prof Henning Sirringhaus, gave me complete freedom to pursue a project of my choosing on this account. I chose to measure Seebeck coefficients within organic transistor channels modulated by a gate voltage and to quantify the extent of disorder within organic polymer systems based on thermoelectric measurements. This article, together with my PhD thesis, was the original seed upon which several projects in organic thermoelectrics were developed within Prof Sirringhaus's group in Cambridge since the year 2014.

5. Deepak Venkateshvaran *et al.*, **Physical Review B 79**, 134405 (2009) **Master Thesis Work**

   This article was the outcome of my master's thesis work undertaken jointly at the Technical University of Munich and the Indian Institute of Technology (IIT) Madras. It became a landmark paper in the field of magnetism; experimentally discovering charge conduction within the so-called 'dirty limit' in magnetic oxides. This work has become the subject matter in textbooks (see Festkörperphysik by Rudolf Gross and Achim Marx). It contributed to understanding the Anomalous Hall Effect (AHE) in Magnetism, documented in Reviews of Modern Physics 82, 1539 (2010).

Top 25 peer-reviewed journal publications in chronology

† joint first authors, * corresponding authors

1. **Deepak Venkateshvaran**, Wolfgang Kaiser, Andrea Boger, Matthias Althammer, M. S. Ramachandra Rao, Sebastian T. B. Goennenwein, Matthias Opel, Rudolf Gross
   Universal scaling behavior of the anomalous Hall effect in $Fe_{3-x}Zn_xO_4$
   Physical Review B 78, 092405 (2008)

2. **Deepak Venkateshvaran**, Matthias Althammer, Andrea Nielson, Stephan Gepraegs, M. S. Ramachandra Rao, Sebastian T. B. Goennenwein, Matthias Opel, Rudolf Gross
   Epitaxial $Fe_{3-x}Zn_xO_4$ thin films: A spintronic material with tunable electrical and magnetic properties
   Physical Review B 79, 134405 (2009)

3. Auke Jisk Kronemeijer, Vincenzo Pecunia, **Deepak Venkateshvaran**, Aditya Sadhanala, John Moriarty, Monika Szumilo, Henning Sirringhaus
   Two-dimensional carrier distribution in top gate polymer field-effect transistors: Correlation between width of density of states and Urbach energy
   Advanced Materials 26, 728 (2014)

4. **Deepak Venkateshvaran**, Auke Jisk Kronemeijer, John Moriarty, David Emin, Henning Sirringhaus
   Field-effect modulated Seebeck coefficient in organic polymers using a microfabricated on-chip architecture
   APL Materials 2, 032102 (2014)

5. **Deepak Venkateshvaran**, Mark Nikolka, Aditya Sadhanala, Vincent Lemaur, Mateusz Zelazny, Michal Kepa, Michael Hurhangee, Auke Jisk Kronemeijer, Vincenzo Pecunia, Iyad Nasrallah, Igor Romanov, Katharina Broch, Ian McCulloch, David Emin, Yoann Olivier, Jerome Cornil, David Beljonne, Henning Sirringhaus
   Approaching disorder-free transport in high mobility conjugated polymers
   Nature 515, 384 (2014)

6. Mathias Gruber, Seok-Heon Jung, Sam Schott, **Deepak Venkateshvaran**, Auke Jisk Kronemeijer, Jens Wenzel Andreasen, Christopher R McNeill, Wallace W. H. Wong, Munazza Shahid, Martin Heeney, Jin-Kyun Lee, Henning Sirringhaus
   Enabling high-mobility, ambipolar charge-transport in a DPP-Benzotriazole copolymer by side-chain engineering
   Chemical Science 6, 6949 (2015)

7. C. N. Warwick, **Deepak Venkateshvaran**, Henning Sirringhaus
   Accurate on-chip measurement of the Seebeck coefficient of high mobility small molecule organic semiconductors
   APL Materials 3, 096104 (2015)

8. Gueorgui O. Nikiforov, **Deepak Venkateshvaran**, Sebastian Mooser, Aurelie Meneau, Thomas Strobel, Auke Kronemeijer, Lang Jiang, Mi Jung Lee, Henning Sirringhaus
   Current-induced Joule heating and electrical field effects in low temperature measurements on TIPS pentacene thin film transistors
   Advanced Electronic Materials 2, 1600163 (2016)

9. Riccardo Di Pietro, Iyad Nasrallah, Joshua Carpenter, Eliot Gann, Lisa Sophie Kölln, Lars Thomsen, **Deepak Venkateshvaran**, Kathryn O'Hara, Aditya Sadhanala, Michael Chabinyc, Christopher R. McNeill, Antonio

Facchetti, Harald Ade, Henning Sirringhaus, Dieter Neher
Coulomb enhanced charge transport in semicrystalline polymer semiconductors
Advanced Functional Materials 26, 8011 (2016)

10. Katharina Broch†*, **Deepak Venkateshvaran**†*, Vincent Lemaur, Yoann Olivier, David Beljonne, Mateusz Zelazny, Iyad Nasrallah, David J Harkin, Martin Statz, Riccardo Di Pietro, Auke Jisk Kronemeijer, Henning Sirringhaus
Measurements of Ambipolar Seebeck Coefficients in High-Mobility Diketopyrrolopyrrole Donor–Acceptor Copolymers
Advanced Electronic Materials 3, 1700225 (2017) – Invited Submission

11. M Statz, **D Venkateshvaran**, X Jiao, S Schott, CR McNeill, D Emin, H Sirringhaus, R Di Pietro
On the manifestation of electron-electron interactions in the thermoelectric response of semicrystalline conjugated polymers with low energetic disorder
Communications Physics 1, 16 (2018)

12. E. Pfitzner, X. Hu, H. W. Schumacher, A. Hoehl, **D. Venkateshvaran**, M. Cubukcu, J.-W. Liao, S. Auffret, J. Heberle, J. Wunderlich, B. Kästner
Near-field magneto-caloritronic nanoscopy on ferromagnetic nanostructures
AIP Advances 8, 125329 (2018)

13. Keehoon Kang, Sam Schott, **Deepak Venkateshvaran**, Katharina Broch, Guillaume Schweicher, David Harkin, Cameron Jellett, Christian Nielsen, Iain McCulloch, Henning Sirringhaus
Investigation of the Thermoelectric Response in Conducting Polymers Doped by Solid-State Diffusion
Materials Today Physics 8, 112 (2019)

14. Shu-Jen Wang†, **Deepak Venkateshvaran**†*, M. R. Mahani, Uday Chopra, Erik R. McNellis, Riccardo Di Pietro, Sam Schott, Angela Wittmann, Guillaume Schweicher, Murat Cubukcu, Keehoon Kang, Remington Carey, Thomas J. Wagner, Janis N. M. Siebrecht, Daniel P. G. H. Wong, Ian E. Jacobs, Razan O. Aboljadayel, Adrian Ionescu, Sergei A. Egorov, Sebasitan Mueller, Olga Zadvorna, Piotr Skalski, Cameron Jellett, Mark Little, Adam Marks, Iain McCulloch, Joerg Wunderlich, Jairo Sinova, Henning Sirringhaus*
Long spin diffusion lengths in doped conjugated polymers due to enhanced exchange coupling
Nature Electronics 2, 98 (2019)

15. Mark Nikolka, Katharina Broch, John Armitage, David Hanifi, Peer J. Nowack, **Deepak Venkateshvaran**, Aditya Sadhanala, Jan Saska, Mark Mascal, SeokHeon Jung, Jin-Kyun Lee, Iain McCulloch, Alberto Salleo, Henning Sirringhaus
High-mobility, trap-free space charge limited currents in low-disorder conjugated polymers enabled by small molecular additives
Nature Communications 10, 2122 (2019)

16. Guillaume Schweicher, Michael T. Ruggiero, Gabriele D'Avino, David J. Harkin, Katharina Broch, **Deepak Venkateshvaran**, Guoming Liu, Audrey Richard, Christian Ruzié, Jeff Armstrong, Alan R. Kennedy, Kenneth Shankland, Kazuo Takimiya, Yves H. Geerts, J. Axel Zeitler, Simone Fratini, Henning Sirringhaus
Accurate, mode-resolved mapping of electron-phonon coupling for the rational design of high mobility molecular semiconductors
Advanced Materials 31, 1902407 (2019)

17. Georg Ulrich, Emanuel Pfitzner, Arne Hoehl, Jung-Wei Liao, Olga Zadvorna, Guillaume Schweicher, Henning Sirringhaus, Joachim Heberle, Bernd Kästner, Jörg Wunderlich, **Deepak Venkateshvaran**

Thermoelectric nanospectroscopy for the imaging of molecular fingerprints
Nanophotonics 9(14), 4347 (2020)

18. Piotr Skalski, Olga Zadvorna, **Deepak Venkateshvaran**, Henning Sirringhaus
Distinguishing spin pumping from spin rectification in lateral spin pumping device architectures based on doped organic semiconductors
Physical Review Materials 6, 024601 (2022)

19. Vishal Panchal, Illia Dobryden, Ude D. Hangen, Dimitrios Simatos, Leszek J. Spalek, Ian E. Jacobs, Guillaume Schweicher, Per M. Claesson, **Deepak Venkateshvaran***
Mechanical Properties of Organic Electronic Polymers on the Nanoscale
Advanced Electronic Materials 8, 2101019 (2022)

20. Illia Dobryden, Vladimir V. Korolkov, Vincent Lemaur, Matthew Waldrip, Hio-Ieng Un, Dimitrios Simatos, Leszek J. Spalek, Oana D. Jurchescu, Yoann Olivier, Per M. Claesson, **Deepak Venkateshvaran***
Dynamic self-stabilization in the electronic and nanomechanical properties of an organic polymer semiconductor
Nature Communications 13, 3076 (2022)

21. **Deepak Venkateshvaran***, Mateo T. R. Cervantes, Leszek J. Spalek, Ki-Hwan Hwang, Kaspars Pudzs, Martins Rutkis, Guillaume Schweicher, Pablo Padilla-Longoria
Understanding the thermoelectric transport properties of organic semiconductors through the perspective of polarons
Advanced Devices and Instrumentation 5, 0067 (2024) – Invited Submission

22. Silvia Cristofaro, Dorothee Brandt, Vincent Lemaur, Ki-Hwan Hwang, Ljiljana Fruk, **Deepak Venkateshvaran**, Luca Muccioli, Silvia Orlandi, Yoann Olivier
Assessing the influence of nanoscale morphology on the mechanical properties of semiconducting polymers
Journal of Materials Chemistry C 13, 15506 (2025) – Invited Submission

23. Ki-Hwan Hwang, Dorothée Brandt, Silvia Cristofaro, Cameron J. Nickerson, Federico Modesti, Mindaugas Gicevicius, Mateo T. R. Cervantes, Martina Volpi, Leszek J. Spalek, Luca Muccioli, Per M. Claesson, Ljiljana Fruk, Yves Geerts, Guillaume Schweicher, Yoann Olivier, Erin R. Johnson, **Deepak Venkateshvaran***
Measuring the molecular origins of stiffness in organic semiconductor thin films
Nature Communications 17, 1621 (2026)

24. Ki-Hwan Hwang, Shun-ichiro Ito, Osnat Z. Arteaga, Mateo T. R. Cervantes, Leszek J. Spalek, Hio-Ieng Un, Guillaume Schweicher, Ljiljana Fruk, Derya Baran, Matthew Smith, Mariano Campoy-Quiles, Taishi Takenobu, **Deepak Venkateshvaran***
Spatially resolved nanomechanics links domain connectivity to charge transport in semiconducting polymers
Under review at Science Advances (2026)

25. Ki-Hwan Hwang, Mateo T. R. Cervantes, Leszek J. Spalek, Yves Geerts, Guillaume Schweicher, Per M. Claesson, Ljiljana Fruk, **Deepak Venkateshvaran***
Direct observation of elastic ripening in molecular crystals
In Preparation (2027)

Complete publications list on Google Scholar: https://scholar.google.co.uk/citations?user=otuUyXIAAAAJ&hl=en